

WBDMR: A Wearable, Biosignal-Driven Music Recommendation System for Emotion

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Abstract

Music therapy has shown great promise in improving mental health, reducing stress and inducing relaxation for patients around the world. Current music therapy requires professional guidance, making it largely inaccessible in the real world. Further, music therapy largely cannot adapt to a person’s emotions outside of consultations. Recently, new methods have emerged utilizing emotion tracking and AI to create recommendations, showing promising results. Otherwise, many patients have utilized self diagnosed music as a replacement. However, these methods each have their own drawbacks. Many emotion tracking and AI models use additional input, including facial photos, manual user input, and chatbot interaction in order to generate recommendations. These interruptions may interfere with the listening experience, and therefore may interfere with the effectiveness of therapy as a whole as well. Moreover, most systems treat emotion as a static input rather than a dynamic target, matching music to either the user’s current input or their desired one, with no transition between the two. These systems are severely lacking in comparison to clinical music therapy as a result, as professional therapy typically follow the Iso-Principle, in which music first reaches the user at their current state before sliding towards a target (Davis, Gfeller, & Thaut, 2008; Altschuler, 1948). User inputted data and self diagnosis may risk user bias as well, as people are often inconsistent in judging their own emotional state, particularly when distressed, and may report what they expect to feel rather than what they actually feel (Barrett et al., 2007). This project presents WBDMR, a closed loop system which utilizes an EmotiBit wearable to capture EDA and PPG biosignals, predict the user’s emotional state on a valence-arousal plane using a Gradient Boosting model trained on the CASE dataset. From there, the system will recommend a gradual playlist transition toward a target emotion from music sourced in the MuSe dataset. The recommendation engine is grounded in the Iso-Principle, where the listener’s current emotional state is first reflected, and steered towards a target throughout the duration of a playlist. The model achieves R^2 scores of 0.119 for valence and 0.187 for arousal on the CASE dataset. <https://musictherapyapp.streamlit.app/>.

1 Introduction

Music therapy has showed consistently positive impacts across a wide range of mental health applications. These include reducing stress and anxiety, emotion regulation in patients with depression, and regulation of chronic pain (Davis, Gfeller, & Thaut 2008). Contrary to popular belief, its effectiveness does not simply stem from the music itself. Instead, it is from the responsive way a trained therapist selects and tweaks the music in real time to match and adjust the patient’s emotional state. This responsiveness is what gives clinical music therapy its edge over simply listening to a personal playlist (Davis, Gfeller, & Thaut 2008). A drawback of music therapy, however, exists in its inaccessibility outside of formal clinical settings. Most formal sessions require a licensed therapist and a controlled setting, placing it out of reach for many who may not be able to lack the time or financial resources to achieve attend such sessions. Further, formal sessions would be unavailable during everyday moments where regulation is needed most, such as managing anxiety during a commute or while trying to decompress after work. To address this, recent work has begun to explore whether technology can help close this gap, building systems that detect a user’s emotional state and recommend music accordingly, without a human therapist or a controlled environment. Koelstra et al (2012) demonstrated that physiological signals such as EDA and heart rate carry emotional information across valence and arousal dimensions. Williams et al. (2020) showed that AI driven music systems show promise for many mental health applications. Given these results, an automated, wearable driven music therapy system is both feasible and clinically motivated.

Existing approaches, however, fall short of clinical music therapy in two key ways. Systems like Kambbham et al’s (2025) facial expression based recommender and Mahmud et al’s CAPeRS (2025) require the user to actively submit a photo or interact with a camera to detect mood. More sophisticated systems, like the CoMEEM recommender, use LLMs and conversational agents to link songs to episodic memories. These input

methods infer emotion and achieve personalization by requesting users to describe memories, identify people and places associated with them, and manually rate their own emotional state before any recommendation is made. Some systems sidestep sensing entirely, with Faisal et al. (2026) inferring emotional state from listening history and self reported mood. These systems fail to reach the effectiveness of clinical music therapy because their input method may cause friction during the extraction period, and interrupt the listening experience. By requesting manual user input, these systems may interrupt the passive, immersive experience that makes music therapeutic in the first place. These methods of describing emotions, face scans, and user input may also be viewed as intrusive, especially for an audience that simply wants to listen. Further, self reported emotional data risks being wholly inaccurate, as people are inconsistent judges of their own emotional state, particularly when distressed, and may simply report what they expect to feel rather than what they actually feel (Barrett et al., 2007). Another limitation exists in the fact that most of these systems do not address emotional transition. In most of the systems available today, the music is directly matched to either the user’s current mood or their current mood. This treats emotion as a fixed input rather than the dynamic trajectory it really is. This principle is fundamentally at odds with what clinical music theory calls the Iso-Principle, a music therapy technique in which the therapist begins by selecting music that matches the client’s current emotional state and gradually shifts towards a target emotion, allowing the listener’s mood to follow the music as opposed to exposing them to an abrupt contrast. Abels & Kasayka (1998) describe it as meeting the client where they are, rather than where you want to go. To this day, no existing recommendation system combines passive biosignal sensing with the Iso-Principle to generate a gradual playlist transition.

This project presents WBDMR, an end to end system that addresses both of these limitations. This project requires no active input for the user, as it uses an EmotiBit sensor to obtain biosignals from the user. It plugs these biosignals into a machine learning pipeline that predicts the user’s current emotional state as coordinates on Russell’s (1980) circumplex model of affect with a machine learning model. Russell’s model envisions a two dimensional valence-arousal space in which valence captures the positivity or negativity of an emotion and arousal captures its intensity. The aforementioned model is a Gradient Boosting Regressor trained on the CASE dataset (Sharma et al., 2019), a dataset that provides continuous valence-arousal annotations alongside ECG and EDA recordings across 30 subjects. Features are then extracted from this dataset using NeuroKit2, with HRV metrics like SDNN, RMSSD, pNN50, and MeanNN, as well as EDA metrics, with its decomposition into phasic and tonic components. From there, the system accepts a user defined target emotion and constructs a playlist using songs sourced from the MuSe dataset of around 90,000 tracks with valence and arousal scores. The system finds points between the detected emotional state and the target to create a rough mathematical approximation of the Iso-Principle. The full pipeline is deployed as a Streamlit web application supporting both a biosignal upload mode and a demo mode, with user inputted emotions.

The project has accomplished a working EmotiBit sensor and file transmission protocol, a working emotion prediction model, and a working recommendation engine. The model achieves R^2 scores of 0.119 for valence and 0.187 for arousal on a test set of 1,094 windows. Although modest, these scores are consistent with what peripheral wrist worn signals can achieve on valence-arousal regression. In addition, this is a substantial improvement over baseline that produced negative R^2 due to a normalization bug, which was identified and corrected during development. It should be noted that this project does not claim to replicate clinical music therapy. Instead, demonstrates that the Iso-Principle can be operationalized in an automated, passive pipeline, and aims to resolve the two main problems concerning current emotion based music therapy systems today.

2 Procedure

All source code is available at <https://github.com/byzhgdot2/MusicTherapyApp>.

2.1 Software

Table 1. Software Components and Details

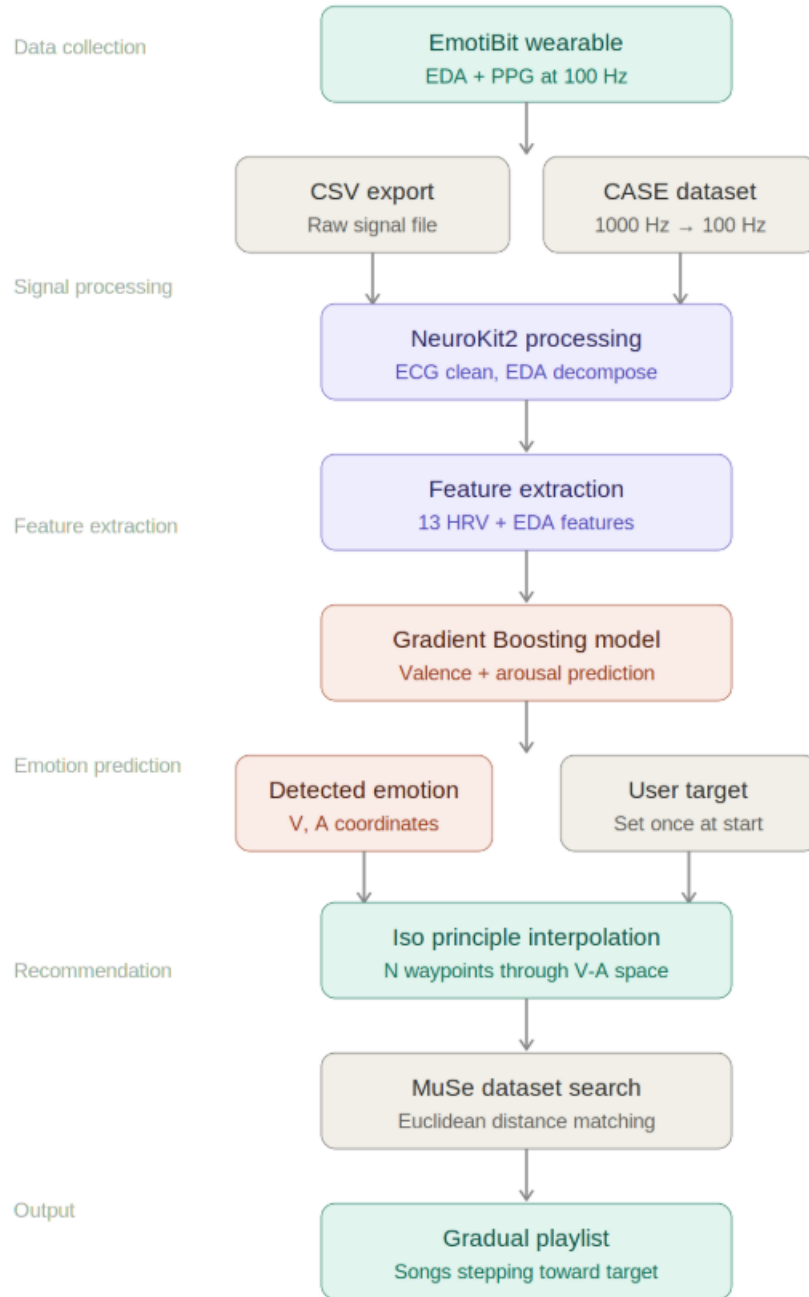


Figure 1: WBDMR Pipeline Overview

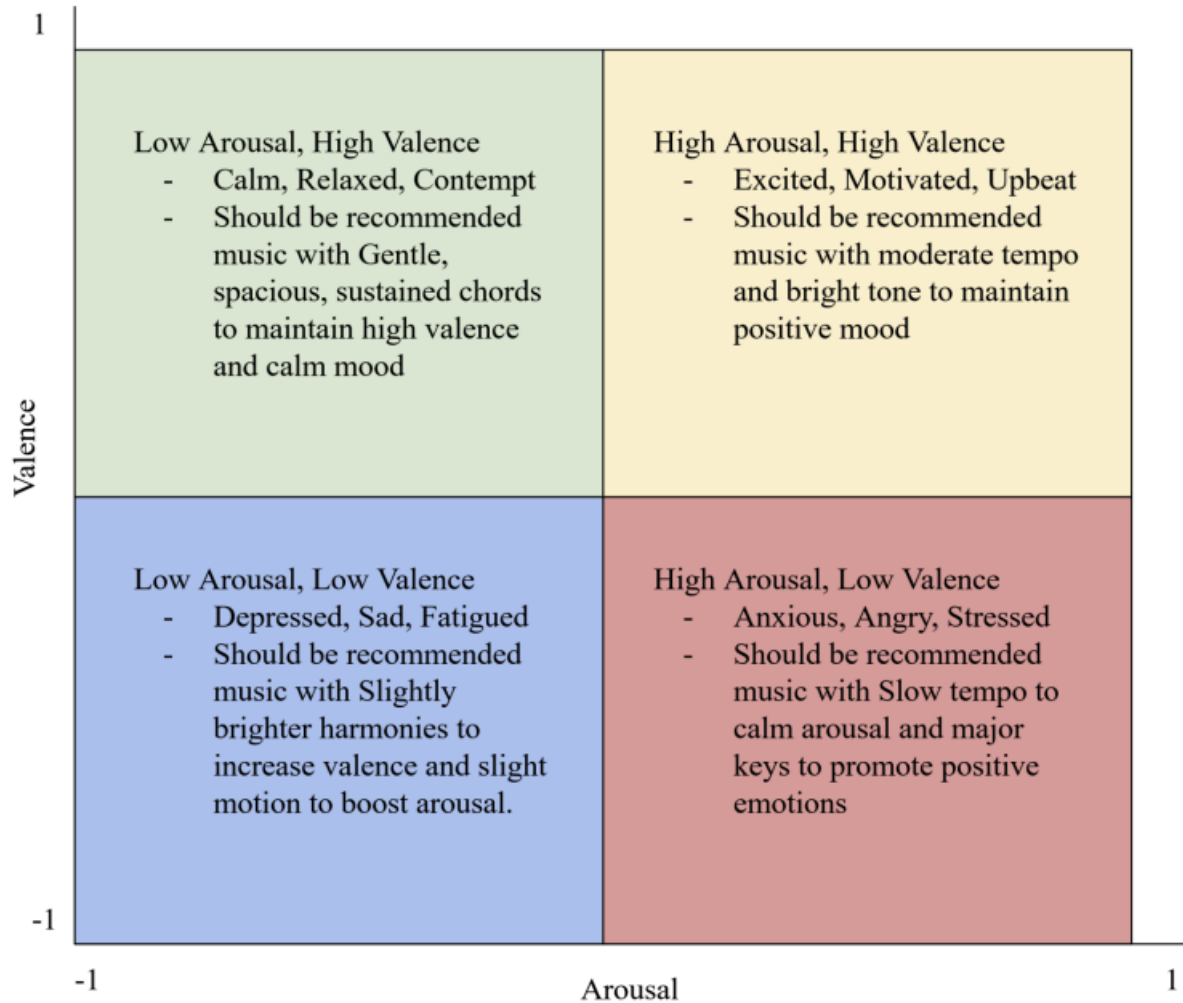


Figure 2: The Valence-Arousal Matrix

Component	Detail
Language	Python 3.14.5
Web Framework	Streamlit
ML Framework	scikit-learn
Key Libraries	NeuroKit2, NumPy, pandas, matplotlib
Datasets	CASE (figshare), MuSe v3

2.2 Hardware / Wearable Device

An EmotiBit MD board was selected for physiological data collection for this project. This was chosen because of its lightweight, untethered characteristics. In addition, it collects EDA and PPG at 100 Hz from the wrist directly. Alternative signals like EEG could theoretically offer richer emotional data, but that may require obstructive headsets that few people would be willing to wear when commuting.

The EmotiBit board was attached to a subject’s wrist using the included elastic wristband. The board measures EDA by applying a constant DC current to the user’s stratum corneum (uppermost layer of skin) (Poh et al., 2010). The difference in voltage from one electrode to another will indicate how resistive the skin is, from which EDA can be derived, usually in microsiemens (). The EmotiBit does this using a silver electrode coated with silver chloride (Ag/AgCl) mounted on the wrist (Montgomery et al., 2024).

The board collects PPG data by shining an infrared, red, and green light onto the skin, which is then reflected into a photosensor. As the volume of blood under the sensor changes, the amount of light reflected by the skin will change, which is read by a photosensor (Kyriacou & Allen, 2021). HR and HRV was calculated from the PPG data using the HeartPy library. The data collected will be wirelessly streamed to a laptop using Wi-Fi and Laboratory Streaming Layer (LSL) for processing.

2.3 Datasets

The CASE dataset (Sharma et al., 2019) was chosen for this project to train the emotion prediction model. Despite having access to the more commonly used DEAP dataset, CASE was chosen due to its usage of ECG and EDA signal over EEG, which was used in the DEAP dataset. The DEAP dataset was also constructed using post-hoc emotion ratings rather than continuous annotations, which may introduce recall bias between the physiological signals and reported emotion. The CASE dataset’s joystick based Emotion Reporting Interface (JERI) captures valence and arousal simultaneously in real time, which maps more cleanly onto a regression target. The dataset covers 30 subjects, 15 male and 15 female, each watching validated video stimuli while physiological signals were recorded at 1000 Hz (Sharma et al., 2019).

The MuSe Dataset was chosen for this project for the recommendation portion of this project. The dataset provides pre computed valence, arousal, and dominance scores for each of its roughly 90,000 tracks, making it a viable lookup table. The Spotify API was the initial candidate for the recommendation system, but Spotify discontinued its valence and energy metadata for third party developers, making it essentially useless for this project. AcousticBrainz, another option, faced the same problem, as it was shut down. MuSe was chosen as a result.

2.4 Signal Processing

Data from the CASE dataset was recorded at 1000 Hz, as opposed to the data collected from the EmotiBit board, at 100 Hz. The data was therefore downsampled to 100 Hz to match the wearable’s output before feature extraction.

The NeuroKit2 library handled ECG cleaning and peak detection, from which HRV metrics were computed. These include SDNN, the standard deviation of all NN intervals; RMSSD, the root mean square of successive differences; pNN50, the percentage of intervals differing by more than 50 ms; and MeanNN. These four metrics were chosen because they are the most established time domain HRV features for emotion recognition in the literature assessed (Task Force, 1996; Liu et al., 2016). In addition, these features can be computed reliably

from windows as short as 30 seconds. This is key to the project given the window size used, given that they were the aforementioned 30 seconds.

As for EDA, NeuroKit2 decomposes them into its phasic and tonic components. The tonic component reflects the slow moving baseline skin conductance level, while the phasic component maps the quick fluctuations which may be tied to specific stimuli or emotional responses. SCR peak count and peak rate were also extracted due to the fact that the frequency of skin conductance responses has been shown to carry arousal information (Boucsein, 2012).

Table 2. Features and Descriptions

Feature	Description
HR_mean	Mean heart rate
HR_std	Standard deviation of heart rate
HRV_SDNN	Standard deviation of all NN intervals
HRV_RMSSD	Root mean square of successive NN interval differences
HRV_pNN50	Percentage of successive NN intervals differing by more than 50 ms
HRV_MeanNN	Mean of all NN intervals
EDA_mean	Mean raw EDA signal
EDA_std	Standard deviation of raw EDA signal
EDA_phasic_mean	Mean phasic skin conductance response
EDA_phasic_std	Standard deviation of the phasic component
EDA_tonic_mean	Mean tonic skin conductance level
EDA_peaks	Total count of SCR peaks
EDA_peak_rate	SCR peaks per second

Other HRV metrics, like LF, HF, and LF/HF ratio were tested and dropped because they required windows of 5 minutes or more to compute reliably. This was far longer than the 30 second segments used in this project. EMG was not collected. Raw signal statistics beyond mean and standard deviation simply added noise without improving R^2 , and were cut. These 13 features were the ones that survived the elimination process.

A median imputer was used for missing values followed by StandardScaler normalization during the preprocessing process.

2.5 Feature Set and Algorithm

After testing all three, Gradient Boosting was chosen over Random Forest and SVM for the machine learning model type. SVM with an RBF kernel failed to generalize across subjects, as the kernel width was sensitive to the feature scale. Ultimately, the cross subject variance proved to be too high. Random Forest performed better but tended to underfit on the arousal dimension, which proved critical especially because the signal tended to be noisier there. Gradient boosting with 200 estimators, a max depth of 3, and a learning rate of 0.05 outperformed both on the validation set, especially for arousal. This shallow tree depth was chosen deliberately to reduce overfitting.

30 second windows with a 10 second step were selected after testing window lengths of 30, 60, and 90 seconds. Despite HRV metrics becoming relatively unreliable below 60 seconds even under ideal conditions, shorter windows with overlap creates significantly more samples per recording. Therefore, by setting windows at 30 seconds, the effective training set was multiplied by three times compared to treating each recording as a single sample. This was a tradeoff that was made to optimize a 30 subject sample, as longer windows produced marginally higher per window accuracy for the tradeoff of fewer training examples, which hurt overall model performance.

Following this, the Gradient Boosting Regressor is wrapped in a MultiOutputRegressor for simultaneous valence and arousal prediction.

2.6 Music Recommendation

The recommendation system relies on a mathematic approximation of the Iso-Principle. Given a current emotion (v_0, a_0), a user defined target emotion (v_t, a_t), and the length of the playlist minus two (N), the system finds N waypoints through the space between the two coordinates of the valence arousal coordinate plane. For each point, the MuSe dataset is searched by distance in the valence arousal plane to find the nearest available song (although this is subject to genre filtering if inputted). As a result, a playlist will be compiled that begins near the listener’s current emotional state, steering gradually towards the target with each step.

Euclidean distance was used for song matching rather than other methods, like cosine similarity or learned embeddings. This was because valence-arousal coordinates in MuSe were already normalized to a comparable scale. The simplicity of Euclidean distance is therefore appropriate here, as there is no strong reason to expect that the geometry of the emotion space is non Euclidean with respect to the resolution this system operates at.

2.7 Experimental Design

The model was evaluated using R^2 , the coefficient of determination, as the primary metric. Because valence and arousal are continuous outputs, classification metrics like accuracy are not appropriate. On the other hand, R^2 directly measures how much of the variance in the target the model explains. Zero means that the model does no better than guessing the mean and 1 means a perfect fit.

Data was split 70/15/15 into train/validation/test sets by way of random sampling of windows. This yielded 7,290 total windows across 30 subjects.

3 Results

3.1 Quantitative Results

Table 4: R^2 scores on the test set.

Output	Test R^2
Valence	0.119
Arousal	0.187

Arousal prediction outperformed valence, which is consistent with the emotion recognition literature reviewed in the process of this project. This is because arousal is more directly reflected in EDA and HRV, with high arousal leading to increases skin conductance and elevating heart rate. Valence, on the other hand, is a more subtle dimension, with physiological difference between happiness and anger, for example, being far less apparent than the difference between excitement and calm (Koelstra et al., 2012; Liu et al., 2016). This asymmetry is a known limitation of physiological signals and is not a specific flaw of this pipeline.

Overall, the R^2 scores achieved are modest. Most EDA/HRV based regression systems on the CASE dataset report similarly limited performance, with the EPiC workshop benchmark on CASE, who used the same dataset with more sophisticated models, achieving correlation coefficients within the range of 0.1 to 0.2 for peripheral signals (Sharma et al., 2019). It should be noted that EEG systems based on the DEAP dataset regularly exceed 80% accuracy, although this can be concluded as a consequence of the richer signal context of EEG and the easier structure of binary classification as opposed to continuous regression.

3.2 Confusion Analysis

Q4 (Calm/Relaxed) is the best performing quadrant with 134 of 214 instances being correctly classified, roughly 63%. This is followed by Q2 (Angry/Tense) at 87 of 158 instances, 55%. This could be attributed to the fact that they are physiologically distinctive, with Q2 involving high sympathetic activation with

elevated skin conductance and compressed HRV, while Q4 involves the opposite, giving the model an easier job in separating these two extremes.

The most confused quadrants is the Q3 (Sad/Depressed) matrix, being predicted as Q4 (Calm/Relaxed) in 257 cases, the largest off diagonal cell by a wide margin. Only in 164 of 582 instances was it correctly identified, around 28% of the time. This could be attributed to the fact that both quadrants 3 and 4 have low arousal. The factor that distinguishes the two is valence, of which peripheral EDA and HRV signals are poorly suited to detect, leading the model to collapse much of Q3 to Q4.

Quadrant 1, (Happy/Excited) was the worst performing quadrant overall. Only 30 of 140 instances had the model correctly predict the result, around 21%, and predictions were scattered across all four quadrants. It should be noted that 64 instances ended with the model predicting Q1 as Q4, which is rather strange given that these two quadrants are literally on different sides of the spectrum. This may reflect the difficulty of estimating valence from wrist worn biosignals, given that Q1 and Q2 overlap heavily in arousal, and Q1 and Q4 overlap heavily with valence, leaving the model without a clean separating signal.

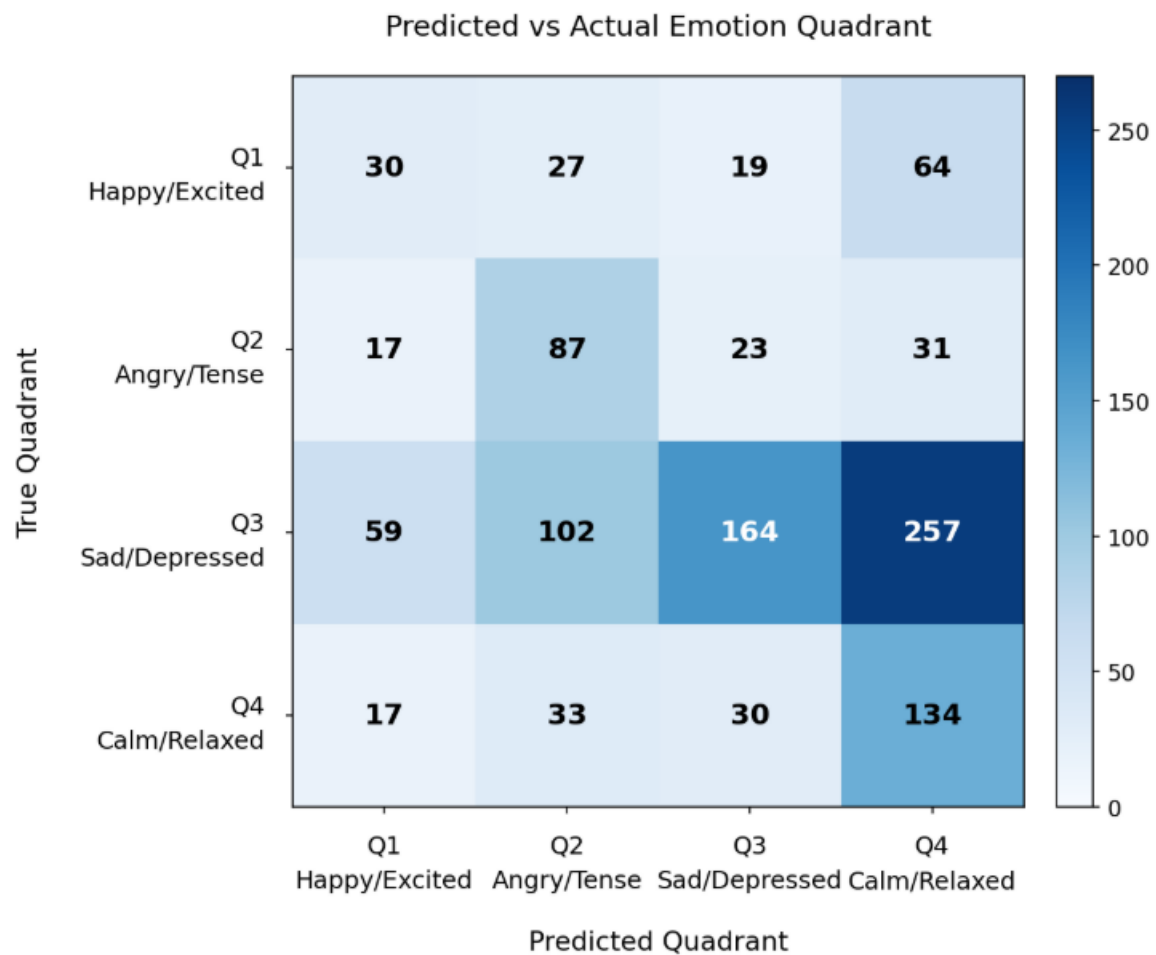


Figure 3: Confusion Matrix

3.3 Training Curves

Depicted in the figure below, train and validation R^2 both can be seen to rise sharply in the first 50 boosting estimators before leveling off, suggesting that beyond 200, adding estimators would yield diminishing returns. A gradual widening of the gap between train and validation R^2 indicates mild overfitting.

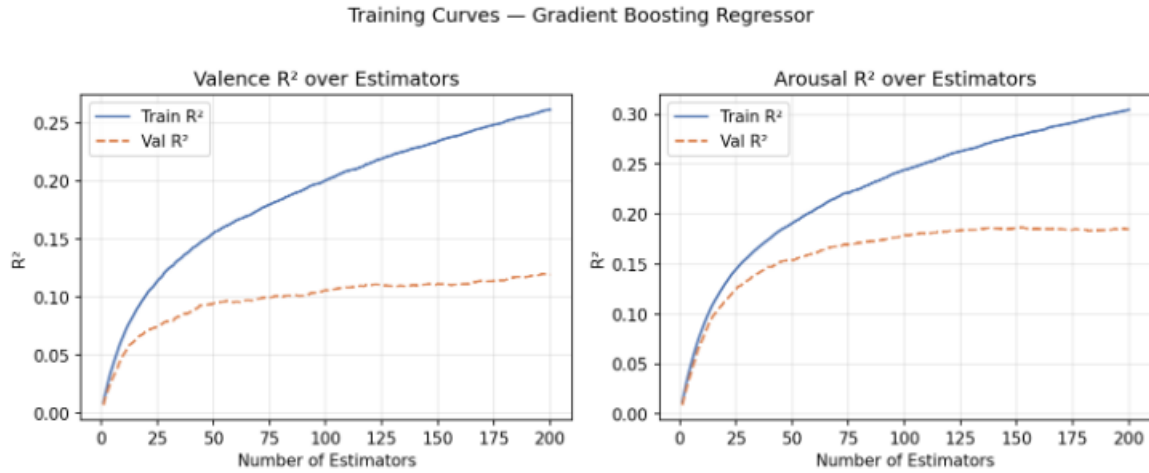


Figure 4: Training curves for Valence and Arousal

3.4 Final Product

All components of this system are operational and integrated. The EmotiBit streams physiological data to a computer. This is exported to CSV and fed into the pipeline. The Streamlit application supports both an Upload Mode which accepts a real physiological CSV from the wearable, which then runs the full prediction and recommendation pipeline. Demo mode allows the user to experience the recommendation engine without the user inputted CSV, so the system can be demonstrated without physiological data.

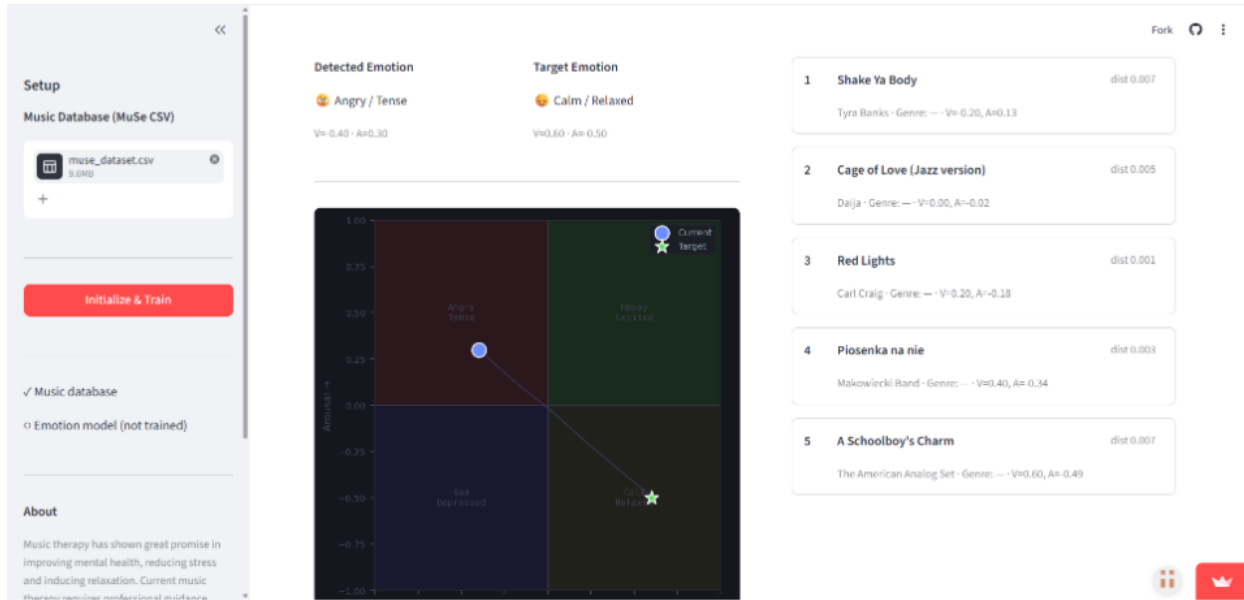


Figure 5: WBDMR Pipeline Overview

4 Conclusions

This project accomplishes a meaningful proof of concept, showing that the Iso-Principle can be operationalized computationally in a passive, wearable driven pipeline that collects signals without any user input, to ultimately steer a user towards their target emotion. No prior automated recommendation system reviewed

does all three of these together, and this project clearly contributes to the field in this way. The Streamlit application is fully deployed and functional in both its Upload Mode and Demo Mode, making the system demonstrable without access to a physical wearable.

Several constraints limit the current system. The model is trained on only 30 subjects, making it difficult to generalize to people it has never seen. This is made even more complicated based on the physiological baselines between individuals. A person could feel a certain way about opposite songs; for example, a person may enjoy and be happier as a result of listening to angry or sad songs, and vice versa. The current project does not address such concerns. The training data itself also introduces another layer of distance from the target case use, as the CASE dataset induced emotion through video stimuli in a controlled lab, as opposed to the everyday stressors this system was meant to address, like commuting or studying. In addition, the CASE dataset’s valence-arousal annotations were created based on a self reporting joystick annotation system, which may introduce user bias to the dataset this model trained on. The Iso-Principle’s implementation in the project was also a simplification. Linear interpolation through a valence-arousal space captures the general idea of gradual transitions, but in clinical therapy, a trained therapist adjusts tempo, instrumentation, and dynamics continuously based on observed patient response, which this system does not in two ways. Firstly, songs are played to completion, and not changed accordingly in response to user response, and the system is captured in a single window, making the user inputted emotional data stale as the playlist goes on. No user study has been run, meaning there is no clear evidence yet that these playlists can actually move a listener towards their target emotional state. What exists right now is a proof of concept as opposed to a validated therapeutic tool.

The R^2 scores achieved, 0.119 for valence and 0.187 for arousal, are modest. Despite this, they sit well within the range of other reported peripheral signal regression systems on CASE. The more meaningful question, which this project does not yet answer, is whether or not a playlist generated by the system actually shifts a listener’s emotional state in its intended direction. This would require that a study that compares Iso-Principle playlists against target only playlists. A natural next step from this project would be that study. When carrying out such a study, considerations, like how to accurately track emotional drift and how to reach such a large array of users, would have to be addressed.

On the technical side, the most valuable near term extension for this project would be eliminating the CSV export step entirely in favor of real time streaming from the EmotiBit. In addition, changing songs that would be recommended next based on the change in a user’s emotions would be another valuable extension. Currently, the system only operates on a single recorded window rather than updating its prediction continuously as the player listens, which means that the emotion estimate goes stale as time goes by. Real time streaming such as this extension would allow the playlist to be live, and adapt as the listener’s physiological state evolves. This would create an even more closed loop architecture than the current project achieves. Longer term, deeper architectures in the form of LSTMs or temporal convolutional networks that can exploit the sequential structure of physiological signals could be able to improve valence prediction, the weaker of the two prediction models that currently exist as a result of this project. The project could also benefit from a larger set of data for the machine learning model to train from.

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